Real Estate Data Analyst Primer

Understanding MLS Data, Real Estate Transactions, and the Analytics Workflow

For analysts with data experience entering the real estate domain

1. Where the Data Comes From: The MLS

The dataset you will be working with originates from CRMLS — the California Regional Multiple Listing Service — one of the largest MLS systems in the United States, covering much of Southern California. Understanding what an MLS is and how it works is the foundation for everything else in this course.

What Is an MLS?

A Multiple Listing Service (MLS) is a private, cooperative database operated by a regional association of real estate brokers. Agents use it to share listing information with each other — essentially a centralized marketplace for properties. There is no single national MLS; instead, each region operates its own (e.g., CRMLS for Southern California, NWMLS for the Pacific Northwest, MRED for the Chicago area).

> 📌 **Analyst Note**
> Consumer-facing sites like Zillow and Redfin pull data from MLS feeds via licensed agreements. When you see a listing on Zillow, you are almost always looking at data that originated in an MLS. Working directly with CRMLS data puts you one step closer to the source than most public datasets.

Key Data Fields You Will Encounter

When a listing agent enters a property into the MLS, they populate a structured form. The fields below are the most important for analysis:

Field Name	Example Value	What It Means
ListingKey	547863987	Unique identifier for the listing record
ListingContractDate	2024-03-01	Date the listing agreement was signed
ListPrice	$875,000	Seller's asking price at time of listing
ClosePrice	$862,500	Final agreed sale price
PurchaseContractDate	2024-03-31	Date the purchase agreement was signed by both parties; typically 30–45 days before CloseDate
CloseDate	2024-04-15	Date the transaction officially closed
LivingArea	1,850	Interior square footage
BedroomsTotal	3	Number of bedrooms

BathroomsTotalInteger	2	Number of bathrooms (integer)
Latitude / Longitude	33.9425 / -118.4	Geographic coordinates for mapping
UnparsedAddress	123 Main St, LA	Full address as entered by the agent

2. The Transaction Lifecycle

Real estate data is not just a snapshot — it captures a sequential business process. Each field in your dataset maps to a specific stage in that process. Analysts who understand this context can avoid misinterpretations that are common when treating the data as abstract rows and columns.

Stage 1 — Listing

The seller signs a listing agreement with a licensed agent. The agent enters the property details into the MLS, creating the listing record. At this point, ListPrice, LivingArea, BedroomsTotal, and address fields are populated. The property is now part of active market supply.

Stage 2 — Buyer Preapproval & Offers

Interested buyers typically obtain a mortgage preapproval before submitting an offer. A preapproval letter from a lender confirms the buyer's borrowing capacity based on income, credit score, and debt-to-income ratio. Sellers often require it before accepting an offer — it functions as a financial qualification filter.

> 💡 **Why This Matters for Analysis**
> Listings that receive many competing offers in a short window often close above list price — a pattern you can detect by comparing ListPrice vs. ClosePrice alongside days-on-market. Conversely, a large gap between list and close price (downward) may signal an overpriced listing or deteriorating market conditions.

Stage 3 — Purchase Agreement & Escrow

When an offer is accepted, both parties sign a purchase agreement that locks in the price, contingencies, and a closing date agreed upon by both buyer and seller — typically 30–45 days out, though this is fully negotiable. That agreed date becomes the CloseDate field in the sold dataset. At this point the property status changes to Pending in the MLS.

Contingencies are conditions that must be met for the deal to proceed. The three most common are:

- Inspection contingency — buyer can cancel or renegotiate if the inspection reveals significant issues
- Appraisal contingency — protects the buyer if the lender's appraisal comes in below the agreed price
- Financing contingency — allows the buyer to exit if their loan falls through

Once the purchase agreement is signed, escrow opens. Escrow is a neutral third-party account managed by a title company or escrow officer that holds the buyer's deposit and all transaction funds until every condition has been met. The closing window allows time for:

- Home inspection — buyer confirms physical condition of the property
- Appraisal — lender's independent confirmation that the property is worth the loan amount
- Loan underwriting — lender's final review of the buyer's financials
- Title search — confirms the seller has clear legal ownership
- Final paperwork — signing, funding, and recording the deed

📌 **When Deals Fall Through**

If a contingency is not satisfied — for example, the inspection reveals major structural damage or the buyer's financing falls through — the deal can be cancelled. When this happens, the property status reverts to Active in the MLS, often labeled Back on Market (BOM). In your data, you may see a property with multiple listing records or status changes. This is not a data error — it reflects a real transaction that failed to close.

Stage 4 — Close

When all contingencies are cleared, escrow closes: funds are transferred, the deed is recorded, and ownership officially changes hands. CloseDate and ClosePrice are then recorded in the MLS. These fields only exist in sold records — active listings that never sell will never have these values, which is a critical structural difference between the listings and sold datasets.

3. List Price vs. Close Price: Why the Difference Matters

This distinction is one of the most important concepts for a real estate data analyst to internalize.

Field Name	Example Value	What It Means
ListPrice	What the seller wants	Marketing price; reflects seller expectations and agent strategy
ClosePrice	What the market will pay	The true transaction value; use this for price trend analysis

The ratio of ClosePrice to ListPrice — often called the sale-to-list ratio — is a key market health indicator. A ratio above 1.0 (close > list) signals a competitive seller's market. This typically occurs when multiple buyers submit competing offers simultaneously, driving the final price above asking. In a bidding war, buyers may also waive contingencies to make their offer more attractive — a detail not captured in the price fields but visible in days-on-market patterns. A ratio below 1.0 signals buyer leverage or an overpriced listing.

📊 **Example Calculation**

ListPrice: $875,000 | ClosePrice: $901,000
Sale-to-List Ratio: $901,000 ÷ $875,000 = 1.030 → sold 3% over asking

Aggregate this metric across ZIP codes and time periods and you have a powerful proxy for local market momentum.

4. Commission Structure and Agent Fields

Real estate commissions are typically 5–6% of the sale price, historically split between the listing agent (seller's side) and the buyer's agent. Since 2024, NAR settlement changes have altered how buyer-agent compensation is structured — buyer-agent fees are now negotiated separately rather than being automatically offered through the MLS.

These agent relationships are captured in your dataset through fields such as:

- Listing agent: ListAgentFirstName, ListAgentLastName, ListOfficeName
- Buyer's agent: BuyerAgentFirstName, BuyerAgentLastName, BuyerOfficeName

For analysts, these fields enable brokerage-level performance analysis — which offices close the most volume, which agents have the highest sale-to-list ratios, and how market share is distributed across brokerages in a given area.

📌 **How Agent Fees Are Actually Paid**
Neither the buyer nor the seller pays agent fees upfront. For the seller, the commission is deducted directly from the sale proceeds at closing — they never write a separate check. For the buyer, the actual payment also happens at closing, not out of pocket beforehand. Critically, commissions are fully contingent on the deal closing. If a transaction falls through, neither agent receives anything. This is a key reason agents are highly motivated to see deals close.

5. Home Financing Basics

Most homebuyers do not pay cash — they finance the purchase with a mortgage. Understanding how mortgages work helps analysts contextualize price data, affordability trends, and buyer behavior in the market.

Down Payment

The down payment is the portion of the purchase price the buyer pays upfront out of pocket. The lender finances the rest. Common down payment amounts are:

- **Low:** 3–5% — minimum for conventional loans (requires private mortgage insurance / PMI)
- **Mid:** 10% — eliminates some PMI requirements depending on the lender

- **Standard:** 20% — the traditional benchmark; eliminates PMI entirely and signals a strong buyer
- **High:** 25%+ — common in investor purchases or jumbo loans

On a $500,000 home, a 20% down payment is $100,000 — leaving a loan balance (principal) of $400,000.

Interest Rate & Loan Term

The interest rate determines how much the buyer pays the lender annually to borrow the principal. The 30-year fixed-rate mortgage is by far the most common loan structure in the U.S. — the rate is locked for the life of the loan and payments are spread evenly over 360 months. Rates fluctuate based on Federal Reserve policy, inflation, and bond markets. As a reference point, rates ranged from roughly 3% (2021) to over 7% (2023–2024).

Monthly Payment Example

The monthly mortgage payment (principal + interest) is calculated using the standard amortization formula. Here is what a $500,000 home looks like at three common rate scenarios, assuming 20% down ($400,000 loan):

Interest Rate	Monthly Payment	Total Interest Paid	Total Cost of Loan
3.0%	$1,686	$207,110	$607,110
6.0%	$2,398	$463,353	$863,353
7.5%	$2,797	$607,024	$1,007,024

💡 **Why This Matters for Analysis**

The monthly payment is what buyers actually qualify for — not the purchase price. When interest rates rise from 3% to 7.5%, the monthly payment on the same $400,000 loan increases by over $1,100. This directly compresses buyer purchasing power and is a primary driver of transaction volume slowdowns. When you see ClosePrice trends flatten or dip in your data, rising rates are often the underlying cause.

6. Listings Dataset vs. Sold Dataset

You will work with two related but structurally distinct datasets:

Field Name	Example Value	What It Means
Listings dataset	All properties entered into MLS	Includes active, expired, withdrawn, and sold listings. Use for supply analysis.
Sold dataset	Properties that completed a	Subset of listings. Always has ClosePrice and CloseDate. Use for price and volume analysis.

	transaction	

7. MLS Status Codes

Every listing in the MLS has a StandardStatus field that tracks where the property is in its lifecycle. This is one of the most important fields for filtering your data correctly — aggregating across statuses without filtering will produce misleading results.

Field Name	Example Value	What It Means
Active	Property is live and available	Use for current inventory and supply analysis. Days on market is actively accumulating.
Pending	Under contract; not yet closed	Offer accepted, purchase agreement signed. Excluded from active supply. ClosePrice not yet set.
Closed	Transaction completed	ClosePrice and CloseDate are populated. Use for price trend and volume analysis.
Back on Market	Returned to Active after going Pending	Deal fell through. Property is available again. Watch for these when analyzing DOM — the clock may have reset.
Expired	Listing term ended without a sale	Seller did not renew with agent. Useful for analyzing overpriced or hard-to-sell properties.
Withdrawn	Seller pulled listing before expiry	Seller chose to exit the market. May indicate a change in seller circumstances.

📌 **Analyst Note**

When building price trend analyses, always filter to StandardStatus = Closed. When measuring active inventory, filter to StandardStatus = Active. Mixing statuses — for example, including Pending records in a price analysis — is a common source of error because Pending records do not yet have a confirmed ClosePrice.

8. Days on Market (DOM)

Days on Market is one of the most widely used metrics in real estate analytics. It measures how long a property has been listed before going under contract or selling. It is a direct indicator of demand — low DOM signals a hot market, high DOM signals soft demand or an overpriced listing.

How It Is Calculated

DOM is typically calculated as the number of calendar days between the ListingContractDate and either the date the status changed to Pending or the CloseDate, depending on the platform. Some MLS systems also track Cumulative Days on Market (CDOM), which adds together DOM across multiple listing periods for the same property — useful for spotting properties that have been relisted after expiring or falling out of contract.

Field Name	Example Value	What It Means
DOM	Days on Market	Days from listing date to Pending or Close for a single listing period
CDOM	Cumulative Days on Market	Total days across all listing periods for the same property; resets only on ownership change

How to Interpret DOM

- **Very Low:** 1–7 days: Extremely competitive — likely received multiple offers quickly
- **Low:** 8–30 days: Healthy demand — normal for most active markets
- **Average:** 31–60 days: Moderate — may reflect seasonal slowdown or slight overpricing
- **High:** 60+ days: Weak demand — property may be overpriced, have condition issues, or face low buyer interest

9. Property Types

Not all residential properties behave the same way in the market or in the data. The PropertyType and PropertySubType fields in the MLS distinguish between different structures, and mixing them in an analysis without filtering can produce misleading results.

Field Name	Example Value	What It Means
Single Family	Detached home on its own lot	The most common property type. Owner controls the land. Typically the highest price per unit. Benchmark for most market analyses.
Condo	Unit within a multi-unit building	Owner holds the interior unit; common areas shared. Subject to HOA fees. Prices per sq ft differ significantly from SFR.
Townhouse	Attached home, usually multi-story	Shares one or two walls with neighbors. May have a small yard. Hybrid between SFR and condo in terms of pricing and ownership.
Multi-Family	2–4 unit residential property	Investor-oriented. Priced on income potential (cap rate) as much as comps. Should be analyzed separately from owner-occupied types.
Manufactured	Factory-built home on leased or owned land	Often lower price points. May not qualify for conventional financing. Separate market dynamics from site-built homes.

📌 **Analyst Note**

Always filter by PropertyType before calculating median price, price per sq ft, or DOM. A ZIP code median that mixes condos and single-family homes will be skewed by the composition of what sold that month — not just by price movement. For most market trend dashboards, Single Family Residential (SFR) is the standard benchmark used by analysts and reported by media.

10. The Analytics Workflow

The work you are doing in this course mirrors a real production workflow used at companies like Zillow, Redfin, CoStar, and regional brokerages. The pipeline has four stages:

1. Raw API Data	JSON / CSV pulls from CRMLS or third-party data vendors. Fields may be inconsistent, have nulls, or require type coercion.
2. Python Cleaning	Standardize field types, handle missing values, deduplicate, apply outlier treatment, and engineer features (e.g., price per sq ft, days on market).
3. Structured Datasets	Clean, analysis-ready tables organized by business object: listings, transactions, agents, offices. These feed downstream tools.
4. Tableau Dashboards	Interactive visualizations: price trend charts, heat maps by ZIP, agent performance leaderboards, days-on-market distributions.

As you move through this course you will execute every stage of this pipeline: pulling raw MLS data, cleaning and structuring it in Python, and connecting the output to Tableau for visualization. By the end, you will have a working, end-to-end analytics workflow — not just a dataset.

11. The Analyst Mindset

Technical skills — SQL, Python, Tableau — are table stakes. What separates a strong analyst from a capable technician is domain knowledge: understanding where the data comes from, what business process generated it, and what questions it can and cannot answer.

As you work through this material, keep asking three questions:

- **Context:** What does this field actually represent in the real world?
- **Skepticism:** Why does this value look the way it does — and when might it be wrong?
- **Purpose:** What decision or insight does answering this question enable?